



Multimodal Digital Twin of a Medieval Armenian Church: 3D Documentation and Acoustic Auralisation

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Abstract

The acoustics of a heritage site is not an accessory to its architecture, but a fragile record of geometry, materials, furnishings and use. Recent work on acoustic heritage has shown the need to document, model and present such spaces as part of cultural heritage, especially where buildings have been altered, damaged or lost (Katz et al., 2020; Martellotta et al., 2008). At the same time, studies of audiovisual acoustic digital twins underline the importance of aligning acoustic measurement points with visual models and immersive presentation systems (Castro & Thompson, 2025). This paper presents a case study of a multimodal 3D and acoustic digital twin of the tenth-century Surb Hovhannes basilica in Byurakan, Armenia. The objective was to test a reproducible workflow that links geometric survey, room acoustic measurement, numerical simulation, spatial auralisation and FAIR-oriented documentation.

The basilica is a stone-built rectangular church with a high interior volume, no dome and a carpeted floor. Its geometry was recorded using a FARO M70 terrestrial laser scanner and Sony A7R III photography. The laser scan data were registered and cleaned in FARO Scene and Leica Cyclone, while the photographic dataset was processed in RealityCapture to produce a textured 3D model. Room impulse responses were measured on site with an IRIS Mini system following ISO 3382-1 practice, and first-order ambisonic recordings of a local choir were used as perceptual reference material. A simplified watertight acoustic model was then prepared in Treble, with explicit assumptions for Armenian tuff masonry, carpets, benches, textiles, openings and the presence of visitors.

The simulation was validated against measured T30, EDT, C80 and Ts values in the 500–2000 Hz range. For three control positions, the average absolute error for T30 was 0.008–0.081 s. This level of agreement supports the use of the model for acoustic reconstruction and spatial auralisation. The validated model was used to generate impulse responses for virtual listening positions and to build an immersive demonstrator in which users can compare dry sound, simulated spatial sound and, where available, measured on-site sound. Figure 1 shows the exterior and interior of the case-study church.

The contribution is relevant to FAIR 3D Heritage because it treats acoustic simulation as a documented extension of 3D heritage data, not as an isolated presentation effect. In line with FAIR principles and paradata-based transparency, the workflow records acquisition settings,

processing steps, source and receiver positions, material assignments, model simplifications and validation metrics (Bentkowska-Kafel et al., 2012; Wilkinson et al., 2016). A CIDOC CRM-compatible metadata layer is proposed to connect the monument, geometric assets, acoustic measurements, material hypotheses, software processes and derived listening experiences (Doerr, 2003). Key challenges remain: long-term storage of heterogeneous files, cataloguing of playable audio and impulse responses, preservation of software-dependent auralisations, and versioning of material assumptions. Nevertheless, the workflow can add immersive listening to existing 3D models and provide a basis for comparing current, reconstructed and hypothetical acoustic states.

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Fig.1. Exterior (left) and interior (right) of the church.



Fig.2. Diagram of the process and its outcomes